Marking Scheme

2026 · Paper 1 · 105 marks

General Rules
- M / A / PROOF mark types
- **f.t.**：承上題錯誤但方法正確者，可獲後續步驟分。
- Deduction cap： A(1)=1 · A(2)=1 · B=0 (u, pp)
- u = 單位遺漏或錯誤；pp = 準確度不符要求。每部分最多扣 1 分。
- 答案正確且步驟合理者可獲滿分。
- 符號書寫略有出入但意思清楚者不扣分。 · Benefit of doubt
- 分數、小數或根式形式均接受，除非題目另有指明。

Section A(1) (35 marks)

1. Change of Subject [B-]

a (3 marks)

M1 $xy = 5y - 3$

M1 $y(x-5) = -3$ f.t.

A1 $y = \dfrac{3}{5-x}$ f.t.

2. Laws of Indices [B]

a (3 marks)

M1 $= \dfrac{m^{-4}n^{6}}{m^{-5}n}$

M1 $= m^{-4-(-5)}n^{6-1}$ f.t.

A1 $= mn^{5}$ f.t.

3. Factorization [B]

a (1 mark)

A1 $(x-2y)(x-3y)$

b (2 marks)

M1 $= (x-2y)(x-3y) + 3(x-3y)$ f.t.

A1 $= (x-3y)(x-2y+3)$ f.t.

4. Applications (Linear Equations) [B+]

a (4 marks)

MLet c concessio tickets be sold; regular $= 4c$ 設優惠券售出c張，普通券$4c$張

M1 $150(4c) + 90c = 44160$ f.t.

M1 $690c = 44160 \Rightarrow c = 64$ f.t.

Atotal $= 5c = 3$ 總數 $= 5c = 320$ f.t.

5. Inequalities [B+]

a (3 marks)

M1 $7 - x \geq 2x - 2 \Rightarrow x \leq 3$

M1 $6 - x < 5 \Rightarrow x > 1$

A1 \(1 f.t.

b (1 mark)

A2 (i.e. $2, 3$) 2（即$2, 3$） f.t.

6. Transformations and Coordinates [B+]

a (2 marks)

A1 $A' = (-3, -4)$

A1 $B' = (8, 8)$

b (2 marks)

Mslope $AB = -$, slope $A'B' =$ 斜率$AB = \frac{-8-3}{8-(-4)} = -\frac{11}{12}$，斜率$A'B' = \frac{8-(-4)}{8-(-3)} = \frac{12}{11}$ f.t.

Aproduct $= -1$, so $AB \perp A'$ 乘積 $= -1$，故$AB \perp A'B'$ f.t.

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7. Pie Charts [B+]

a (2 marks)

Mspring angle $= \frac{1}{5} \times 3$ 春季扇形角 $= \frac{1}{5} \times 360° = 72°$

A1 $x = 360° - 72° - 100° - 80° = 108°$ f.t.

b (2 marks)

M $\frac{108°}{360°} \times ($ $\frac{108°}{360°} \times (\text{總數}) = 180$ f.t.

Atotal $= 600$ 總數 $= 600$ f.t.

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8. Variation [B+]

a (2 marks)

M $y = \frac{k}{\sqrt{x}}$; $k = 972$ $y = \frac{k}{\sqrt{x}}$；$81 = \frac{k}{12} \Rightarrow k = 972$

A1 $y = \frac{972}{\sqrt{x}}$ f.t.

b (3 marks)

M $x = 324$: $y = \frac{972}{18}$ $x = 324$: $y = \frac{972}{18} = 54$ f.t.

Mchange $= 54 - 8$ 變化 $= 54 - 81$ f.t.

A$= -27$ (a decrease of 27) $= -27$（減少27） f.t.

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9. Accuracy of Measurement [B+]

a (2 marks)

Mmaximu absolute error = 5 mL 最大絕對誤差 $= \frac{10}{2} = 5mL$

A1 最小容量 $= 200 - 5 = 195mL$ f.t.

b (3 marks)

Mtotal range $= [23.4, 2$ L 120瓶總容量範圍 $= [120 \times 195,\ 120 \times 205) = [23400, 24600)mL = [23.4, 24.6)L$ f.t.

Mmeasur 23.3 L $\Rightarrow$ total $\in [23.25,$ L 量度為23.3L表示總容量 $\in [23.25, 23.35)L$ f.t.

Athe two ranges are disjoint, so disagree 兩範圍不相交，故不同意 f.t.

Section A(2) (35 marks)

10. Circles and Sectors [A]

a (2 marks)

PROOF $OP = OR(given); PS = RS(Smid - point); OS = OS(common) \Rightarrow \triangle OPS \cong \triangle ORS(SSS).$

A2 $correctproof$

b (4 marks)

M $\angle POQ =$ $\angle POQ = 2\angle PRQ = 20°$（圓心角 $= 2 \times$ 圓周角）

Mby symmetry $\angle POR =$ 由對稱$\angle POR = 2\angle POQ = 40°$（$OQ$平分$\angle POR$） f.t.

Msector area $= \frac{40°}{360°}$ 扇形面積 $= \frac{40°}{360°} \times \pi(6)^2$ f.t.

A1 $= 4\pi cm^2$ f.t.

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11. Statistical Measures [A-]

a (5 marks)

M13 data; median is the 7th value 數據共13個，第7個為中位數

Amedian $= 71$ 中位數 $= 71$ f.t.

Mstandar deviation $= \sqrt{\frac{1}{13}}$ 標準差 $= \sqrt{\frac{1}{13}\sum(x_i - 70)^2}$

A1 $= 8$ f.t.

accept 8.00

A(units $) （單位\)

b (2 marks)

M7 workers have wage > 70 時薪 > 70的有7人

A1 $P = \frac{7}{13}$ f.t.

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12. Similar Solids [A]

a (3 marks)

Mprism volume $= 1680$; volume ratio $= 8 : 27$ 柱體體積 $= 84 \times 20 = 1680cm^3$；體積比 $= (4 : 9)^{\frac{3}{2}} = 8 : 27$

Mlarger $= \frac{27}{35} \times$ 較大錐體 $= \frac{27}{35} \times 1680$ f.t.

A1 $= 1296cm^3$ f.t.

b (4 marks)

Mbase $= 324$, side $= 18$ for larger $1296 = \frac{1}{3} \times (底) \times 12 \Rightarrow 底 = 324$，邊長 $= 18$ f.t.

Msmaller side $= 12$, height $= 8$ 較小錐體邊長 $= 18 \times \frac{2}{3} = 12$，高 $= 12 \times \frac{2}{3} = 8$ f.t.

Mslant $= 10$; TSA $= 12^2 +$ 斜高 $= \sqrt{8^2 + 6^2} = 10$；總表面積 $= 12^2 + 4 \times \frac{1}{2} \times 12 \times 10$ f.t.

A1 $= 384cm^2$ f.t.

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13. Equations of Circles [A]

a (2 marks)

Mradius $= GE =$ | 半徑 $= GE = \sqrt{(-6-2)^2+(5+1)^2} = 10$

A1 | $C:\ (x-2)^2+(y+1)^2=100$ f.t.

b (2 marks)

M $GF = \sqrt{}$ | $GF = \sqrt{(3-2)^2+(11+1)^2} = \sqrt{145}$

A $\sqrt{145} >$, so F lies outside C | $\sqrt{145} > 10$，故F在C外 f.t.

c (3 marks)

A(i) F, G, H are collinear with G between F and H | $(i)F$、G、H共線且G在F與H之間

M(ii) slope $FG = 12$ | (ii)斜率$FG = \dfrac{11-(-1)}{3-2} = 12$ f.t.

A1 | $(ii)y = 12x - 25$ f.t.

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14. Division of Polynomials [A+]

a (3 marks)

Mcompar coefficien of x^2 | $f(x) = (2x^2+ax+4)(3x+7)+bx+c$；比較$x^2$係數：$-13 = 6\cdot ? + 3a$

Mexpandi $3a+14 =$ | 展開比較：$3a+14=-13$ f.t.

A1 | $a=-9$ f.t.

b (5 marks)

PROOF | $(i)f(x) = (2x^2+ax+4)Q_1+(bx+c) and g(x) = (2x^2+ax+4)Q_2+(bx+c); subtract: f-g = (2x^2+ax+4)(Q_1-Q_2), divisible.$

A2 | $(i) correct proof$

M(ii) roots of $2x^2-9x$: $x = \dfrac{9\pm}{4}$ | $(ii)f(x)-g(x)=0$的根為$2x^2-9x+4=0$的根：$x = \dfrac{9\pm 7}{4}$ f.t.

M(ii) $x=4$ or $x=\dfrac{1}{2}$ | $(ii)x=4$或$x=\dfrac{1}{2}$ f.t.

A(ii) $\dfrac{1}{2}$ is not an integer, so disagree | $(ii)\dfrac{1}{2}$非整數，故不同意 f.t.

Section B (35 marks)

15. Logarithmic Functions [A]

a (4 marks)

M $0 = a+1$ and $3 = a+1$ | $0 = a+\log_b 4$及$3 = a+\log_b 32$

Msubtrac $3 = \log_b$ | 相減：$3=\log_b 8 \Rightarrow b=2$ f.t.

M $a=-2$, so $y=-2+$ | $a = -\log_2 4 = -2$，故$y=-2+\log_2 x$ f.t.

A1 | $x = 2^{y+2}$ f.t.

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16. Geometric Sequences [A]

a (2 marks)

M1 $S_{20} = \frac{1.8 \times 10^7 (1 - 0.9^{20})}{1 - 0.9}$

A1 $\approx 1.58 \times 10^8 m^3$ f.t.

r.t. 1.58e8

b (2 marks)

M1 $S_\infty = \frac{1.8 \times 10^7}{1 - 0.9} = 1.8 \times 10^8$

A 1.8×10^8 , so agree

$1.8 \times 10^8 < 1.9 \times 10^8$，故同意 f.t.

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17. Probability [A]

a (2 marks)

M1 $\frac{C_4^4 C_{16}^1}{C_{20}^5}$

A1 $= \frac{1}{969}$ f.t.

b (2 marks)

M1 $\frac{C_4^3 C_{16}^2}{C_{20}^5}$

A1 $= \frac{10}{323}$ f.t.

c (2 marks)

M $1 - P(4$

$1 - P(4 \text{ green}) - P(3 \text{ green})$ f.t.

A1 $= \frac{938}{969}$ f.t.

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18. Quadratic Equations and Discriminant [A+]

a (3 marks)

M1 $x^2 + (2 - 2k)x + (-3k - 11) = 0$

M1 $\Delta = (2 - 2k)^2 - 4(-3k - 11) = 4k^2 + 4k + 48 = 4(k^2 + k + 12)$ f.t.

A $1 - 48 <$, hence two distinct points

$k^2 + k + 12$的判別式 $= 1 - 48 < 0$且首項為正 $\Rightarrow \Delta > 0$，故有兩相異交點 f.t.

b (5 marks)

M(i) $\alpha + \beta =$

$(i) \alpha + \beta = 2k - 2,\ \alpha\beta = -3k - 11$

M1 $(i)(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = (2k - 2)^2 - 4(-3k - 11)$ f.t.

A1 $(i) = 4k^2 + 4k + 48$ f.t.

M(ii) $|AB|^2 =$

$(ii)|AB|^2 = (\alpha - \beta)^2 = 4k^2 + 4k + 48 = 4(k + \frac{1}{2})^2 + 47 \geq 47$ f.t.

A(ii) $|AB| \geq \sqrt{}$, so it is not possible

$(ii)|AB| \geq \sqrt{47} > 4$，故不可能小於4 f.t.

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19. Trigonometry in 3-D [A+]

a (2 marks)

M $\frac{AC}{\sin 108^\circ}$ — $\angle ABC = 180^\circ - 30^\circ - 42^\circ = 108^\circ$；$\frac{AC}{\sin 108^\circ} = \frac{24}{\sin 30^\circ}$

A1 — $AC \approx 45.7cm$ f.t.

r.t. 45.7

b (11 marks)

M(i) heights fall linearly; $\frac{CF}{AC} = \frac{2}{8}$ — (i)高由A（10）至C（2）沿AC線性下降；$\frac{CF}{AC} = \frac{2}{10-2} = \frac{1}{4}$

A1 — $(i) CF = \frac{45.7}{4} \approx 11.4cm$ f.t.

r.t. 11.4

M(ii) $AB \approx 32$; $AF \approx 57$ — $(ii) AB = \frac{24 \sin 42^\circ}{\sin 30^\circ} \approx 32.1$；$AF = AC + CF \approx 57.1$ f.t.

M(ii) $[\triangle ABF]$ — $(ii)[\triangle ABF] = \frac{1}{2} AB \cdot AF \sin 30^\circ$ f.t.

A1 — $(ii) \approx 458cm^2$ f.t.

r.t. 458

M(iii) plane ABC meets ground along BF; use normals — (iii)平面ABC與地面交於BF；由法向量得傾角 f.t.

A(iii) inclination $\approx 21.4^\circ$ — (iii)傾角 $\approx 21.4^\circ$ f.t.

r.t. 21.4

M(iv) B, D, F on ground; compute $[\triangle BDF]$ by coordinat — $(iv) D$、B、F於地面；$[\triangle BDF]$由坐標計算 f.t.

M(iv) $[\triangle BDF]$ cm^2 — $(iv)[\triangle BDF] \approx 427cm^2$ f.t.

A(iv) $427 < 46$, so disagree — $(iv) 427 < 460$，故不同意 f.t.